



WEB PHISHING DETECTION

A PROJECT REPORT

Submitted by

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CHAPTER 1

INTRODUCTION

1.1 PROJECT OVERVIEW:

Now a days, Internet Usage is backbone for a daily routine of our life. In this Scenarios, Various Web Applications needs some adequate and personal information to give a personalize features and information to the users to built a growth of own organizations and profit. Hence, User needs to give some information about them, and made to be easy work, which makes very difficult and needed one. In some Areas, Attackers uses their tactical and trap moves to steal the personal information of the user without knowing the knowledge of the user.

In this Project, we have to develop a detection application with the help of the Domain “Applied Data Science” along with “Machine Learning” by using “Flask Framework” in Python.

1.2 PURPOSE:

Phishing is a type of Social Engineering where an attacker sends a fraudulent (e.g., spoofed, fake, or otherwise deceptive) message designed to trick a person into revealing Sensitive information to the attacker or to deploy malicious software on the victim's infrastructure like ransomware. Phishing attacks have become increasingly sophisticated and often transparently mirror the site being targeted, allowing the attacker to observe everything while the victim is navigating the site, and transverse any additional security boundaries with the victim. As of 2020, phishing is by far the most common attack performed by cybercriminals, the FBI’s Internet Complaint Centre recording over twice as many incidents of phishing than any other type of computer crime. Attempts to prevent or mitigate the impact of phishing incidents include Legislation, user training, public awareness, and technical security measures. Phishing awareness has become important at home and at the work place. For instance, from 2017 to 2020, phishing attacks have increased from 72% to 86% among businesses.

In Modern world, Many Technologies are built and create to crack such Strategy. In This project, we have to discuss and creating a Detection tool to detect and aware from such type of sites.

CHAPTER 2

LITERATURE SURVEY

2.1 EXISTING PROBLEM:

By using existing web Application, it has an initial stage of the project and some of small number of tools only available to create such type of project makes sometimes accuracy is less than the prediction. Numerous numbers Researches and Development to create such type of projects to detect such type of sites by pasting URL. In these projects have unwanted errors and predicted problems makes accuracy related issues that makes very uncomfortable for using such sites. It makes requirement of new Technology or domain which makes this application more secure and accuracy based one to trust by using the user.

2.2 REFERENCES:

1.Sahingo, O. K., Buber, E., Demir, O. and Diri, B. (2019). Machine learning based phishing detection from URLs. Expert System Applications.(117), (pp. 345–357). doi: 10.1016/j.eswa.2018.09.029.

2.Jain A.K., Gupta B.B. “PHISH-SAFE: URL Features-Based Phishing Detection System Using Machine Learning”, Cyber Security. Advances in Intelligent Systems and Computing, vol. 729, 2018, https://doi.org/10.1007/978-981-10-8536-9_44

3.Hung Le, Quang Pham, Doyen Sahoo, and Steven C.H. Hoi, “URLNet: Learning a URL Representation with Deep Learning for Malicious URL Detection”, Conference’17, Washington, DC, USA, arXiv:1802.03162, July 2017.

4.Wu CY, Kuo CC, Yang CS,” A phishing detection system based on machine learning” In: 2019 International Conference on Intelligent Computing and its Emerging Applications (ICEA), pp 28–32, 2019.

5.Parekh, Shraddha & Parikh, Dhwanil & Kotak, Srushti & Sankhe, Prof. (2018). A New Method for Detection of Phishing Websites: URL Detection. 949-952. 10.1109/ICICCT.2018.8473085.

6. Hassan Y.A. and Abdelfettah B, —Using case- based reasoning for phishing detection”, Procedia Computer Science, vol. 109, 2017, pp. 281–288.,

View Article ,Google Scholar.

7. Rao RS, Pais AR. Jail-Phish: An improved search engine based phishing detection system. *Computers & Security*. 2019 Jun 1;83:246–67.

8.Aljofey A, Jiang Q, Qu Q, Huang M, Niyigena JP. An effective phishing detection model based on character level convolutional neural network from URL. *Electronics*. 2020 Sep;9(9):1514.

9.Aleroud A, Karabatis G. Bypassing detection of URL-based phishing attacks using generative adversarial deep neural networks. In: *Proceedings of the Sixth International Workshop on Security and Privacy Analytics* 2020 Mar 16 (pp. 53–60).

10.J. Anirudha and P. Tanuja,||Phishing Attack Detection using Feature Selection Techniques —, *Proceedings of International Conference on Communication and Information Processing (ICCIP)*, 2019, <http://dx.doi.org/10.2139/ssrn.3418542>

2.3 PROBLEM STATEMENT DEFINITION:

Phishing is the most commonly used social engineering and cyber-attack. Through such attacks, the phisher targets naïve online users by tricking them into revealing confidential information, with the purpose of using it fraudulently. Phishing is a form of fraud in which the attacker tries to steal sensitive information such as login credentials or account information by sending as a reputable entity or person in email or other communication channels. There are a number of users who purchase products online and make payments through e-banking. There are e-banking websites that ask users to provide sensitive data such as username, password & credit card details, etc., often for malicious reasons. This type of e-banking website is known as a phishing website. Web phishing aims to steal private information. It will lead to information disclosure and damage to reputation. Large organizations may get trapped in different kinds of scams. Web phishing is one of many security threats to web services on the Internet. This project mainly focuses on applying a machine-learning algorithm to detect phishing websites.

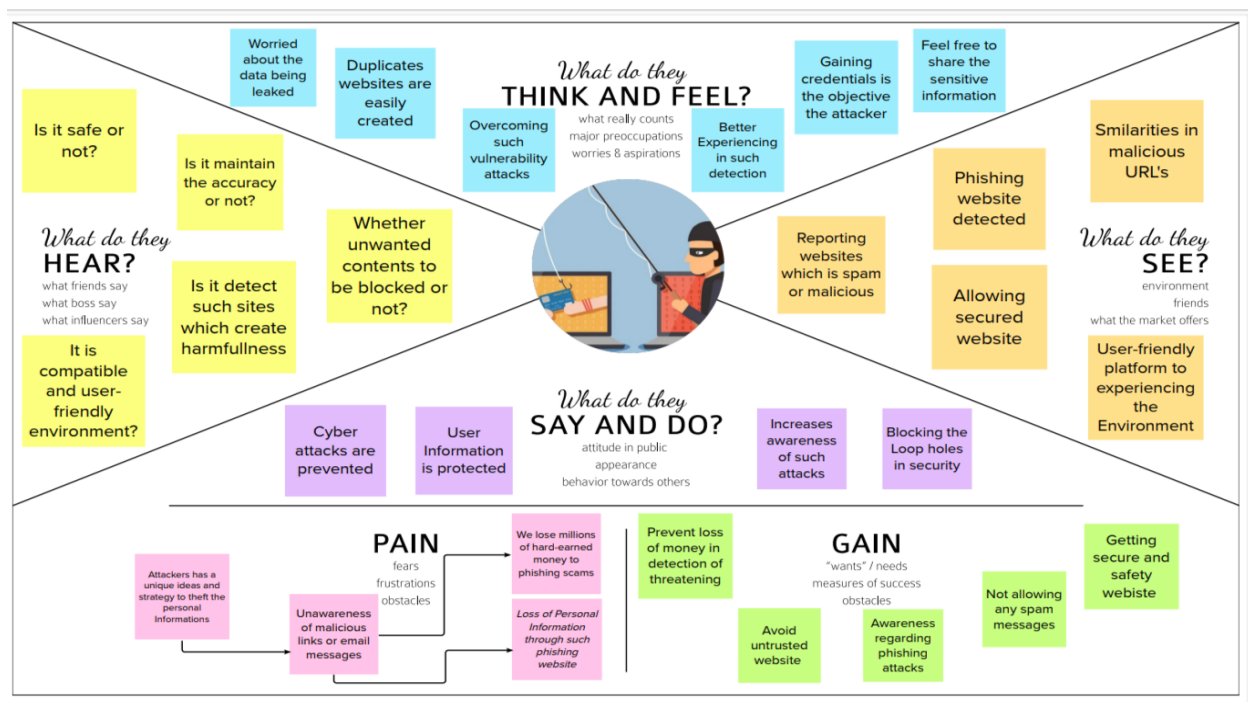
CHAPTER 3

IDEATION & PROPOSED SOLUTION

3.1 EMPATHY MAP CANVAS

An empathy map is a collaborative visualization used to articulate what we know about a particular type of user. It externalizes knowledge about users in order to

- 1) create a shared understanding of user needs, and
- 2) aid in decision making.



Empathy Map Canvas

Traditional empathy maps are split into 4 quadrants (Says, Thinks, Does, and Feels), with the user or persona in the middle. Empathy maps provide a glance into who a user is as a whole and are not chronological or sequential. Empathy maps should be used throughout any UX process to establish common ground among team members and to understand and prioritize user needs. In user-centred design, empathy maps are best used from the very beginning of the design process.

3.2 IDEATION & BRAINSTORMING

Ideation is often closely related to the practice of brainstorming, a specific technique that is utilized to generate new ideas. A principal difference between ideation and brainstorming is that ideation is commonly more thought of as being an individual pursuit, while brainstorming is almost always a group activity. Brainstorming is usually conducted by getting a group of people together to come up with either general new ideas or ideas for solving a specific problem or dealing with a specific situation. For example, a major corporation that recently learned it is the object of a major lawsuit may want to gather together top executives for a brainstorming session on how to publicly respond to the lawsuit being filed.

Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

1. 10 minutes to prepare
2. 1 hour to collaborate
3. 15 minutes to summarize

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

1. 10 minutes

Brainstorming rules

Brainstorming is a creative process in the session and not a linear process of brainstorming or a linear process.

Before you brainstorm

Brainstorming is a creative process in the session and not a linear process of brainstorming or a linear process.

Define your problem statement

Show the top 3 brainstorm questions that you created and the group brainstorm ideas to help in identifying a question to move forward by choosing what problem you are going to solve. Place your problem in a clear, right the statement. This will be the focus of your brainstorm.

1. 10 minutes

Project Flow

1. Problem: The Main Theme or Issue of What Problem is to be solved

2. Problem: For what problem does the idea exist? For what problem does the idea exist?

3. Problem: Make the safe and secure society and Environment by introducing planning

Brainstorm solo

For each member to have an idea that related to the problem statement which is a problem that comes to mind and that can address your problem statement. It is very useful for understanding the possible ways to get an idea which is related to the problem statement.

1. 10 minutes

Ideation from Members

1. Don't visit unrelated sites such as after and eating this

2. Don't visit unrelated sites such as after and eating this

3. Don't visit unrelated sites such as after and eating this

4. Don't visit unrelated sites such as after and eating this

5. Don't visit unrelated sites such as after and eating this

6. Don't visit unrelated sites such as after and eating this

7. Don't visit unrelated sites such as after and eating this

8. Don't visit unrelated sites such as after and eating this

9. Don't visit unrelated sites such as after and eating this

10. Don't visit unrelated sites such as after and eating this

Brainstorm as a group

Have everyone move their ideas into the "group sharing space" within the template and have the team identify and group them. As a team, sort and group them by theme/topic or categories. Discuss and answer any questions that arise. Brainstorming "Yes, and..." and build on the ideas of other people along the way.

1. 10 minutes

Achieve the Goal

Give each person a 5-minute session in which they should present their ideas on.

1. 10 minutes

Flow of Ideas

1. Problem: The Main Theme or Issue of What Problem is to be solved

2. Problem: For what problem does the idea exist? For what problem does the idea exist?

3. Problem: Make the safe and secure society and Environment by introducing planning

Ideation Process

1. Problem: The Main Theme or Issue of What Problem is to be solved

2. Problem: For what problem does the idea exist? For what problem does the idea exist?

3. Problem: Make the safe and secure society and Environment by introducing planning

NEVER GIVE UP

Ideation & Brainstorming

3.3 PROPOSED SOLUTION:

Having hooked your audience into the problem, now you want to paint a picture of what the world will be like when you solve the problem. Your proposed solution should relate the current situation to a desired result and describe the benefits that will accrue when the desired result is achieved. So, begin your proposed solution by briefly describing this desired result. 14 Note: now that you have described the final result, think of your iterative enhancement plan as showing your audience the steps by which you will lead them to that result.

S.No.	Parameter	Description
1.	ProblemStatement (Problem to be solved)	● Phishing is a fraudulent technique that is used over the internet to steal a user's personal information. ● Many cyber infiltrations are accomplished through phishing attacks, where users are tricked into interacting with web pages that appear to be legitimate. ● Nowadays, there are many people who are purchasing products and making transactions online through various websites. ● Sensitive data like passwords, credit card details and bank account information are asked by malicious websites to steal user's information. So, these kinds of malicious activities should be detected.

2.	Idea / Solution description	● Machine learning algorithms can be used to determine the difference between legitimate websites and phishing websites. ● Phishing websites can be detected based on features extracted from the URL.
3.	Novelty/ Uniqueness	● The user will be notified if the website is malicious, such that their privacy will be ensured and awareness will be created. ● The browser extension for web phishing detection helps the user not need to detect a phishing website because the browser extension alerts the user if they enter the phishing website.
4.	Social Impact / Customer Satisfaction	● Phishing has a list of negative effects such as loss of money, loss of intellectual property, damage to reputation, and disruption of operational activities. The technology of phishing detection is constantly being updated. ● Users will come to know whether their details are safe or not. By using the web phishing detection website, the user can check their websites by copying and pasting the URL. Depending on the result, they can be completely safe or not from the above-mentioned impacts. They are safe

		if it is a legitimate URL. Otherwise, they are not safe if it is a phishing URL.
5.	Business Model (Revenue Model)	● A revenue model is a framework for generating financial income. ● There is a scope for including Advertisements in the website such that revenue can be generated. ● There are two plans, with one being "free" and the other being "premium." When classifying a website, the free plan would include the advertisements. The premium plan is a monthly or yearly that eliminates all the advertisements.
6.	Scalability of the Solution	● There are several machine learning algorithms out there. So, we choose the algorithm that gives more accurate results than other algorithms. This makes the user feel safe and secure when making an online purchase using e-payments.

3.4 PROBLEM SOLUTION FIT:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem.

Problem-Solution fit canvas 2.0		Purpose / Vision	
Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? ➤ The customer which uses the website or any third-party platform. ➤ User which can use a banking website or any personal information application and email-users. ➤ Customer which gets stuck and trap in a spam or malicious website.	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices. ➤ By installing some firewalls and anti-virus websites to prevent this type of problems. ➤ Adding the extensions to the browsers such as web phishing detection tool to detect malicious website. ➤ Don't trust online gaming websites which is also trap for users.	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking ➤ By using such detection tool to aware such type of detection without visiting or bring personal information to the attacker. ➤ Avoid trusting rumours about the phishing sites which is one of the strategy to stuck in the trap. ➤ Keep safe in the internet world.
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one, explore different sides. ➤ The usage of such web site which can trap the existing users by send spam messages or getting warning for the spoilage of money. ➤ By using such website, insecure one can getting users sensitive information for the purpose of getting money or spam of malicious website. ➤ Getting information or advertisement from unknown senders.	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job?  Flow of back story behind in the internet	
3. TRIGGERS TR What triggers customers to act? i.e. seeing their neighbour installing Install trusted and safe application without getting backward of malicious spam or virus attack. 4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? User do what are said by the attackers like losing the money or getting stuck with personal information.	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. ➤ We provide tool to detect such type of threat by adding extension or application which is created by using HTML, python and web framework named as a Flask. ➤ This website is more user-friendly, free-environment to explore what is what. By using such type of phishing detection tool to detect the phishing site and keep safe and secure from internet world.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 Don't click suspicious website to avoid traps of attackers through email or messages, which prevents such type of actions 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. Avoid click offline messages such as online games and money earning websites to take safe & secure	

Problem Solution Fit

CHAPTER 4

REQUIREMENT ANALYSIS

4.1 FUNCTIONAL REQUIREMENT

Functional requirements are the desired operations of a program, or system as defined in software development and systems engineering. The systems in systems engineering can be either software electronic hardware or combination software-driven electronic

- Address Bar based Features.
- Abnormal Based Features
- HTML and JavaScript Based Features.
- Domain Based Features.

FR No.	FunctionalRequirement (Epic)	SubRequirement(Story/Sub Task)
FR-1	User Input	User can enter the URL in respective areato verify its genuinely.
FR-2	Detection toolcomparison	Model checksand verify the websites usingMachine learning algorithm.
FR-3	Features Extraction	After using the Algorithm, it found whether the website is legitimate website or Phishing website.

FR-4	Prediction	Model predicts the URL using Machine Learning Algorithms such as ● Logistic Regression ● Random Forest Regression / Classification ● Decision Tree Regression / Classification ● K-Nearest Neighbors ● Support Vector Machine
FR-5	Results	Model send all the output to the classifier to produce the Final result.
FR-6	Actions	This Model displays the malicious sites with accuracy and reliability.

4.2 NON-FUNCTIONAL REQUIREMENT

Non-functional Requirements (NFRs) define system attributes such as security, reliability, performance, maintainability, scalability, and usability. They serve as constraints or restrictions on the design of the system across the different backlogs.

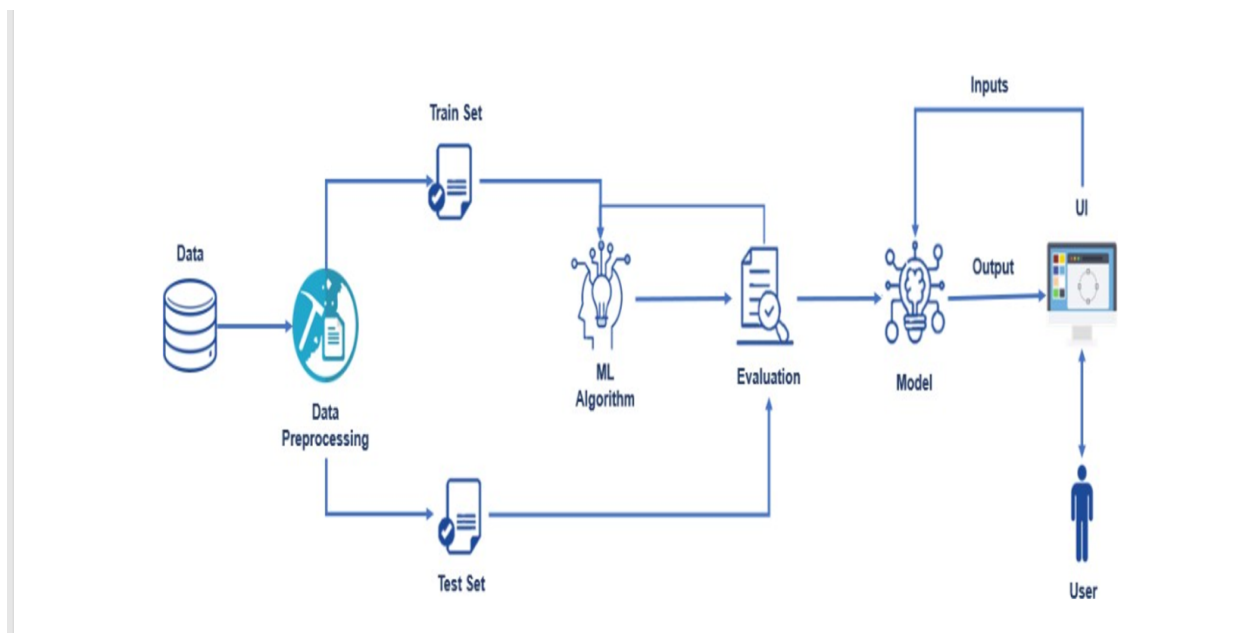
FR No.	Non-Functional Requirement	Description
NFR-1	Usability	User can experience user-friendly environment to explore everything
NFR-2	Security	User can work without fear about the Security
NFR-3	Reliability	It can be accuracy because of Machine learning
NFR-4	Performance	It can be performed efficiently by detecting those attacks and vulnerability
NFR-5	Availability	It may be working in expected outcomes
NFR-6	Scalability	There is a number of testing and number of parameters, thus creates scalability for user

CHAPTER 5

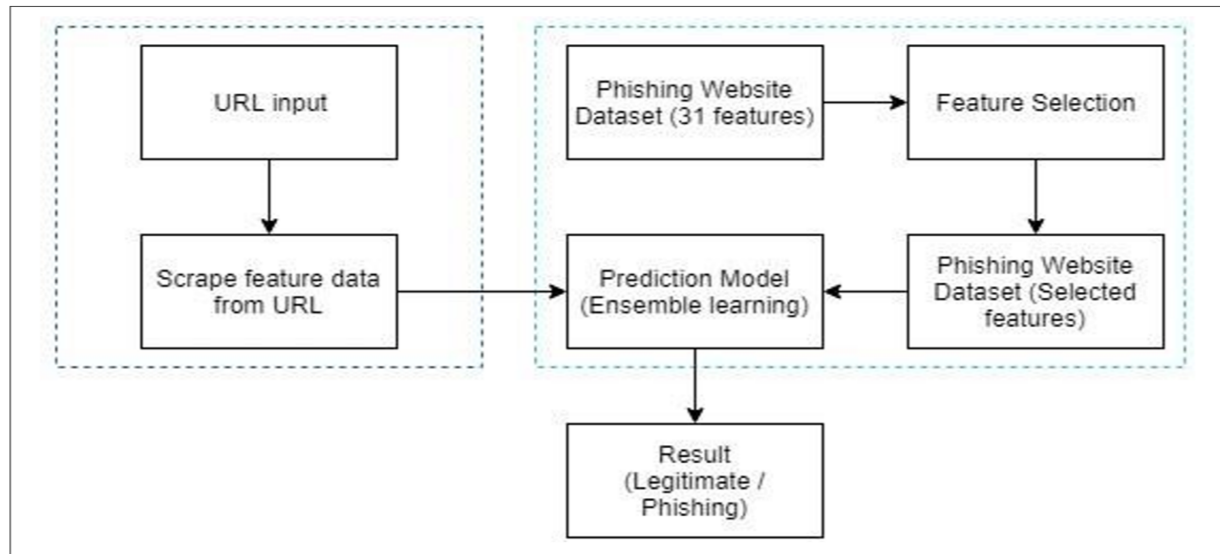
PROJECT DESIGN

5.1 DATA FLOW DIAGRAMS

A data flow diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows, plus short text labels, to show data inputs, outputs, storage points and the routes between each destination. Data flowcharts can range from simple, even hand-drawn process overviews, to in-depth, multi-level DFDs that dig progressively deeper into how the data is handled. They can be used to analyse an existing system or model a new one. Like all the best diagrams and charts, a DFD can often visually —say— things that would be hard to explain in words, and they work for both technical and nontechnical audiences, from developer to CEO. That’s why DFDs remain so popular after all these years



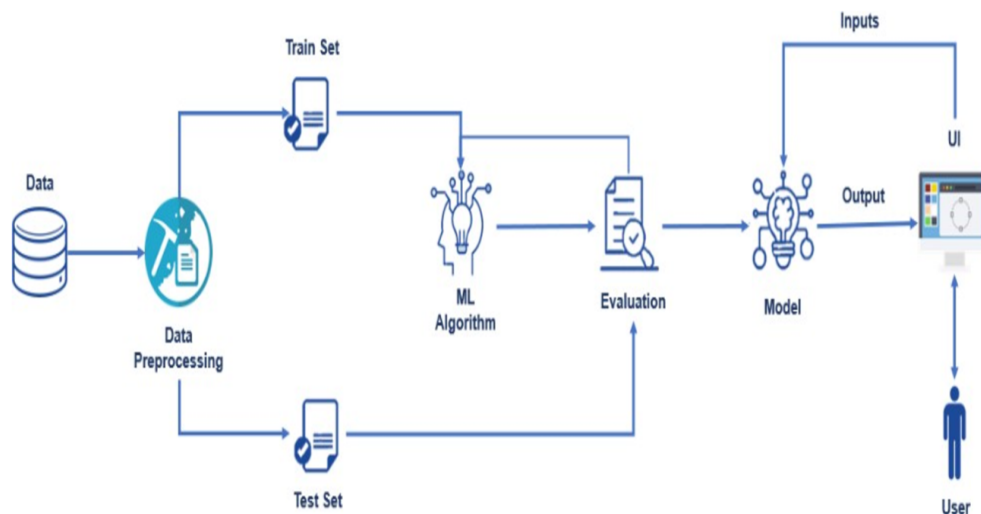
Data Flow Diagram



5.2 SOLUTION & TECHNICAL ARCHITECTURE

This project will be approached based on decision tree, random forest and SVM algorithms.

- The data is collected from the dataset and it is pre-processed. Then it is applied to the machine learning algorithmic model and analyzed.
- The model is trained and tested according to the problem specified. Python Flask and IBM cloud facility is used in developing this project.



5.3 USER STORIES

A user story is the smallest unit of work in an agile framework. It's an end goal, not a feature, expressed from the software user's perspective. A user story is an informal, general explanation of a software feature written from the perspective of the end user or customer. The purpose of a user story is to articulate how a piece of work will deliver a particular value back to the customer.

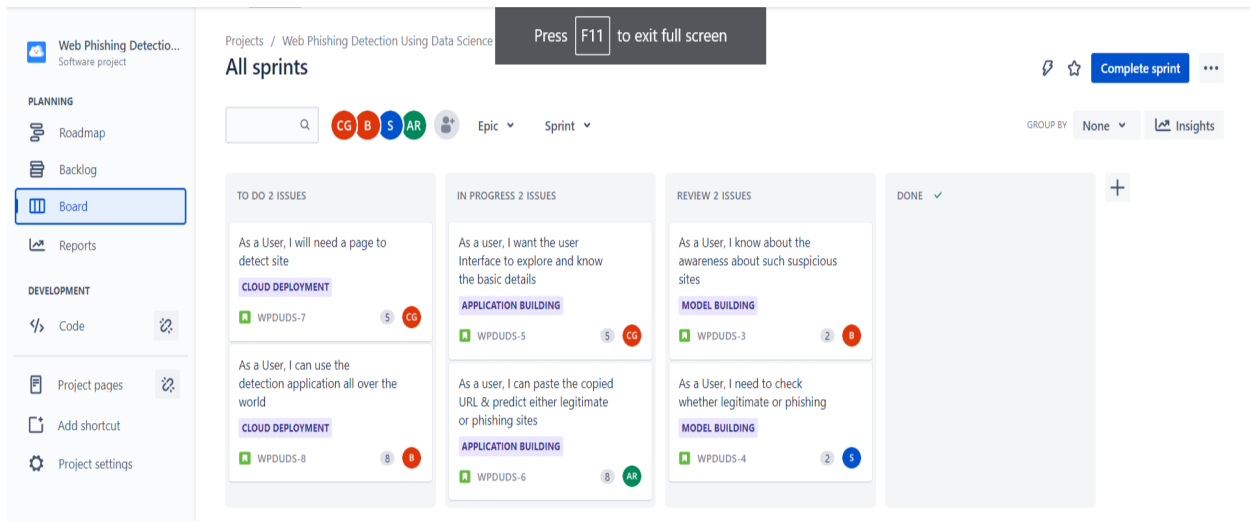
User Type	Functional Requirement (Epic)	User Story Number	User Story/Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Home Page	USN-1	As a user,I can use this user-friendly application without any registration and login	I can access thishome page	High	Sprint-2
	Detectpage	USN-2	As a user, I can copy or enterthe URL in the searchbox to detect the site	I can accessthis detectpage	High	Sprint-2

	Solution	USN-3	As a user, I can get the solution with the help of Machine learning by trained model	I can access this solution page	Medium	Sprint-3
Customer (WebUser)	Home Page	USN-1	As a user, I can use this user-friendly application without any registration and login	I can access this homepage	High	Sprint-2
	Detect Page	USN-2	As a user, I can copy or enter the URL in the searchbox to detect the site	I can access this detectpage	High	Sprint-2
	Solution	USN-3	As a user, I can get the solution with the help of ML	I can access this solution page	Medium	Sprint-3

CHAPTER 6

PROJECT PLANNING & SCHEDULING

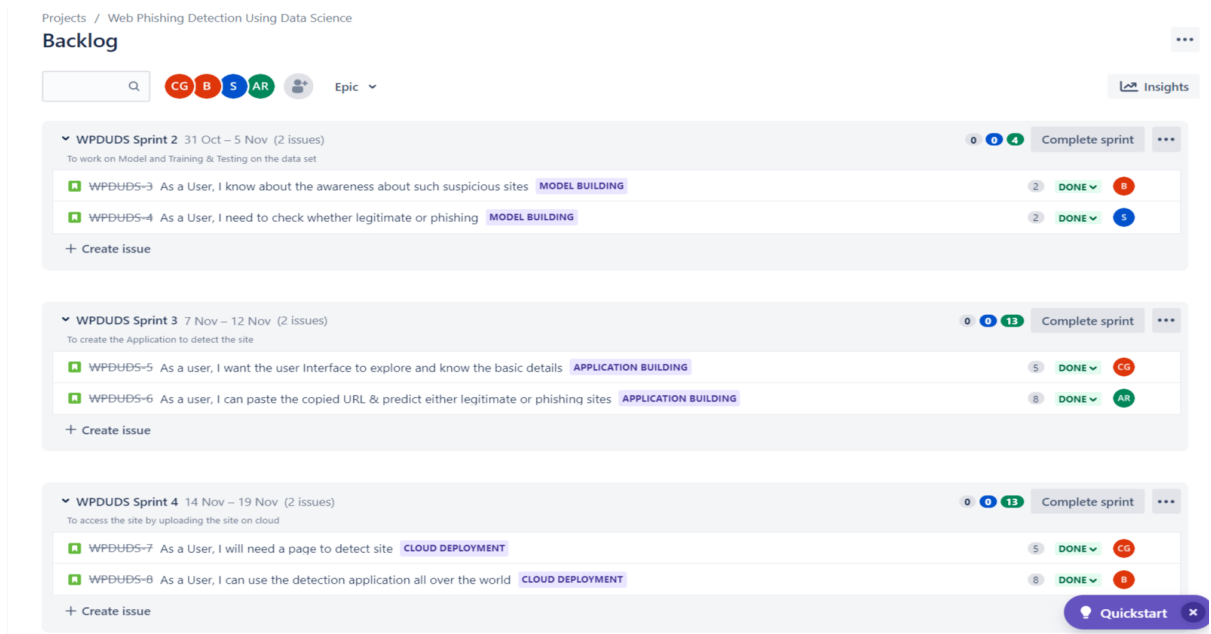
As per an IBM project, Project Planning & Scheduling phase conducting to track and know the current stage of the project or intimating a remaining tasks and milestones to achieve the Goal of the Project. In this planning phase contains a Sprint Planning & Estimation and Delivery Schedule and also completion of Sprint and task in JIRA Software to generating a report of the project.



Project Planning & Sprint Creation

6.1 SPRINT PLANNING & ESTIMATION

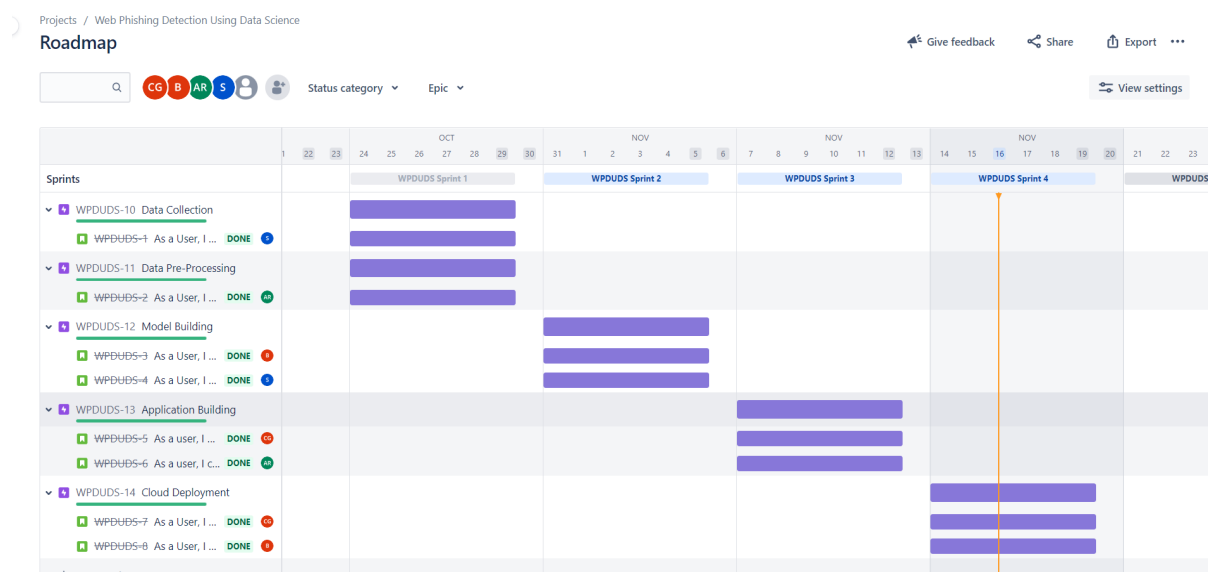
In the Milestone Activities task, Team leader assigns a corresponding Task(epic) to the team members, and give a time interval to complete the tasks. Leader takes a responsibility to collect the results of the respective tasks. Based on the project's task leader plan the activities and segregate the tasks to distribute to individual one to complete the corresponding sprint which is detailed in the milestone activities.



Sprint Planning & Estimation

6.2 SPRINT DELIVERY SCHEDULE

Sprint Completion should be taken to one day before of deadline. Sprint Delivery from the team members must be submit in before timeline of the Submit date. Scheduling of the task to the members plays an important role to build the application(project) development faster and quicker completion of the work.



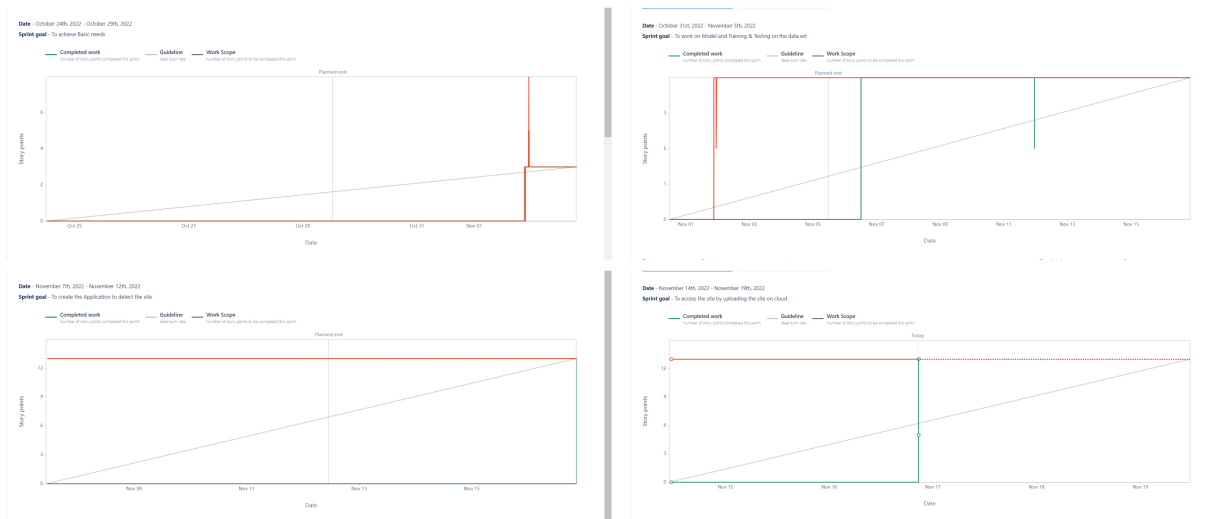
Sprint Delivery Schedule

6.3 REPORTS FROM JIRA

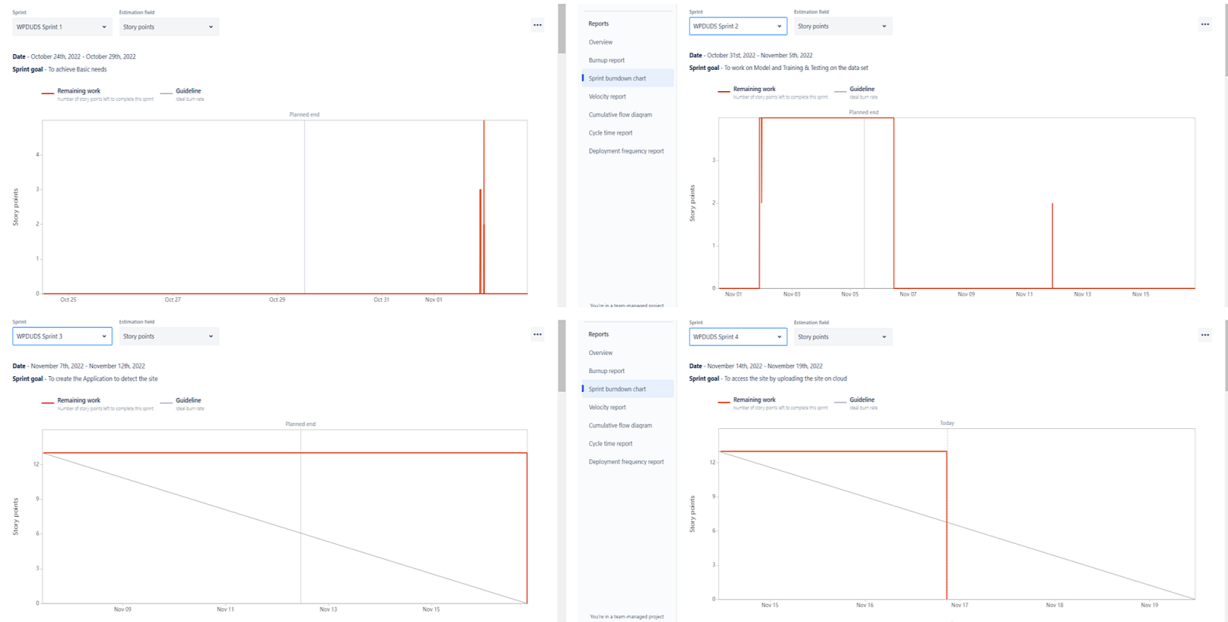
The JIRA Software, which is used to reporting the tasks and activities assigning to the team members, to generate the results of the assigning tasks. This Software contains various facilitates to indicate the project process flow and assigning tasks on the members. It features Roadmap, Kanban Board and Reports such as Burndown chart, velocity chart, Burnup chart, etc.,

It initializes a backlog of the task which indicates the starting phase or initial stage of sprint or phase that makes in progress by the team leader. The completion of the project take move on the Kanban board and at last it gets done status on the Kanban board. After the completion of the tasks in the sprint, Complete sprint status given by the leader to remove the sprint of the backlog.

REPORTS:



1. Burnup Chart



2.Burndown Chart



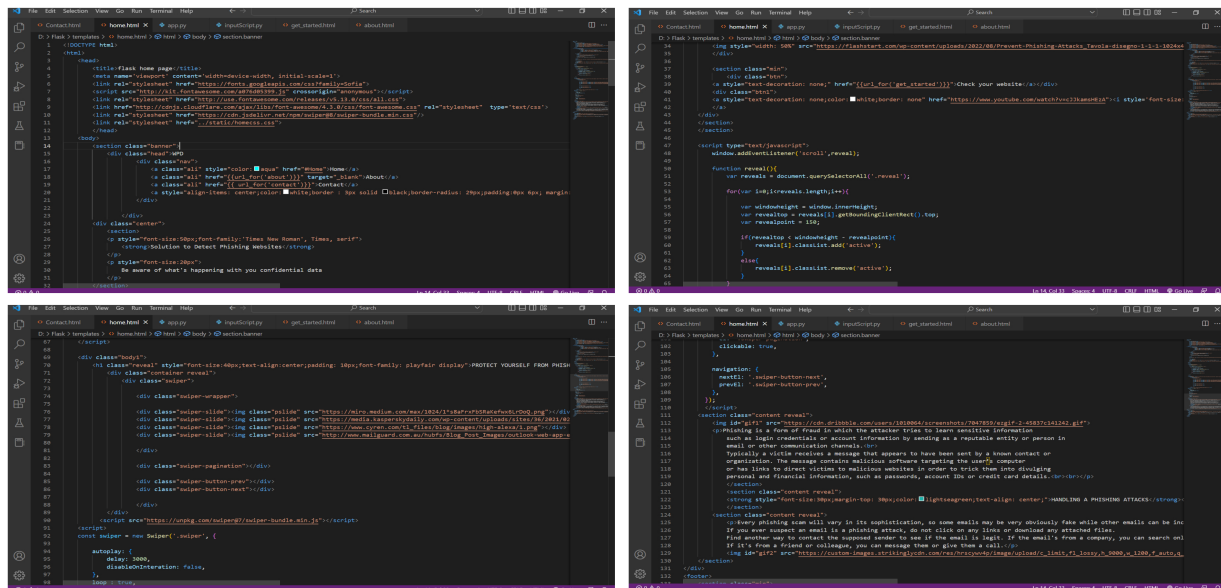
3.Velocity Report

CHAPTER 7

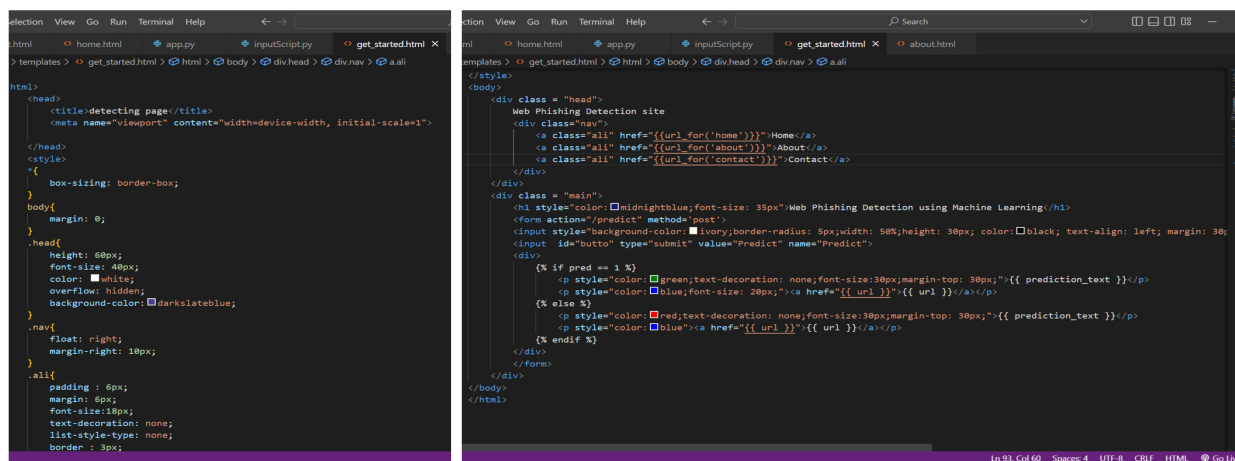
CODING & SOLUTIONING

7.1 FEATURE 1

User Interface or Front-End of the Project is given below as a Screenshot. It contains home, contact, predict and about pages which is linking and redirected to various pages for user convenience.



1.Home Page.html



2.Get_started.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <meta charset="UTF-8">
5 <title>Web Phishing Detection site</title>
6 </head>
7 <body>
8 <div class="nav">
9 <a href="#">Home</a>
10 <a href="#">About</a>
11 <a href="#">Contact</a>
12 </div>
13 <div class="about">
14 <h1>Web Phishing Detection site</h1>
15 <p>Web Phishing is one of many security threats to web services on the Internet.
16 Web phishing aims to steal private information, such as usernames, passwords, and credit card
17 details, by way of impersonating a legitimate entity.
18 </p>
19 <p>Web Phishing is one of many security threats to web services on the Internet.
20 Web phishing aims to steal private information, such as usernames, passwords, and credit card
21 details, by way of impersonating a legitimate entity.
22 </p>
23 </div>
24 <div class="contact">
25 <h2>Contact Us</h2>
26 <form>
27 <input type="text" value="" placeholder="Name" name="name" required>
28 <input type="email" value="" placeholder="Email" name="email" required>
29 <input type="text" value="" placeholder="Phone" name="phone" required>
30 <input type="text" value="" placeholder="Address" name="address" required>
31 <input type="text" value="" placeholder="City" name="city" required>
32 <input type="text" value="" placeholder="State" name="state" required>
33 <input type="text" value="" placeholder="Zip" name="zip" required>
34 <input type="text" value="" placeholder="Subject" name="subject" required>
35 <input type="text" value="" placeholder="Message" name="message" required>
36 <input type="button" value="Send" />
37 </form>
38 </div>
39 </body>
40 </html>
```

```
100 align-items: center;
101 font-size: 20px;
102 padding-top: 20px;
103 height: 100px;
104 color: #000000;
105 display: flex;
106 align-content: center;
107 flex-direction: column;
108 justify-content: center;
109 font-family: Georgia, 'Times New Roman', Times, serif;
110 }
111 .nav {
112 margin: auto;
113 align-content: center;
114 }
115 .about {
116 padding: 10px;
117 align-items: center;
118 font-size: 18px;
119 }
120 .contact {
121 padding: 10px;
122 align-items: center;
123 font-size: 18px;
124 }
```

```
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18 </p>
19 <p>Web Phishing is one of many security threats to web services on the Internet.
20 Web phishing aims to steal private information, such as usernames, passwords, and credit card
21 details, by way of impersonating a legitimate entity.
22 </p>
23 </div>
24 <div class="contact">
25 <h2>Contact Us</h2>
26 <form>
27 <input type="text" value="" placeholder="Name" name="name" required>
28 <input type="email" value="" placeholder="Email" name="email" required>
29 <input type="text" value="" placeholder="Phone" name="phone" required>
30 <input type="text" value="" placeholder="Address" name="address" required>
31 <input type="text" value="" placeholder="City" name="city" required>
32 <input type="text" value="" placeholder="State" name="state" required>
33 <input type="text" value="" placeholder="Zip" name="zip" required>
34 <input type="text" value="" placeholder="Subject" name="subject" required>
35 <input type="text" value="" placeholder="Message" name="message" required>
36 <input type="button" value="Send" />
37 </form>
38 </div>
39 </body>
40 </html>
```

```
100 Web Phishing Detection site
101 <div class="nav">
102 <a class="all" href="#{{url_for('home')}}">Home</a>
103 <a class="all" href="#{{url_for('about')}}">About</a>
104 <a class="all" href="#{{url_for('contact')}}">Contact</a>
105 </div>
106 <div class="about">
107 <h1>ABOUT</h1>
108 <div class="content">
109 <p>Web services is one of the key communications software services for the Internet.
110 Web phishing is one of many security threats to web services on the Internet.
111 Web phishing aims to steal private information, such as usernames, passwords, and credit card
112 details, by way of impersonating a legitimate entity.
113 </p>
114 <p>Web Phishing is one of many security threats to web services on the Internet.
115 Web phishing aims to steal private information, such as usernames, passwords, and credit card
116 details, by way of impersonating a legitimate entity.
117 </p>
118 </div>
119 </div>
```

3.About Page.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <meta charset="UTF-8">
5 <title>Web Phishing Detection site</title>
6 </head>
7 <body>
8 <div class="nav">
9 <a href="#">Home</a>
10 <a href="#">About</a>
11 <a href="#">Contact</a>
12 </div>
13 <div class="about">
14 <h1>Web Phishing Detection site</h1>
15 <p>Web Phishing is one of many security threats to web services on the Internet.
16 Web phishing aims to steal private information, such as usernames, passwords, and credit card
17 details, by way of impersonating a legitimate entity.
18 </p>
19 <p>Web Phishing is one of many security threats to web services on the Internet.
20 Web phishing aims to steal private information, such as usernames, passwords, and credit card
21 details, by way of impersonating a legitimate entity.
22 </p>
23 </div>
24 <div class="contact">
25 <h2>Contact Us</h2>
26 <form>
27 <input type="text" value="" placeholder="Name" name="name" required>
28 <input type="email" value="" placeholder="Email" name="email" required>
29 <input type="text" value="" placeholder="Phone" name="phone" required>
30 <input type="text" value="" placeholder="Address" name="address" required>
31 <input type="text" value="" placeholder="City" name="city" required>
32 <input type="text" value="" placeholder="State" name="state" required>
33 <input type="text" value="" placeholder="Zip" name="zip" required>
34 <input type="text" value="" placeholder="Subject" name="subject" required>
35 <input type="text" value="" placeholder="Message" name="message" required>
36 <input type="button" value="Send" />
37 </form>
38 </div>
39 </body>
40 </html>
```

```
200 <div class="contact">
201 <h2>CONTACT</h2>
202 <div class="contact-wrapper">
203 <div class="form-group">
204 <input type="text" value="" placeholder="Name" name="name" required>
205 </div>
206 <div class="form-group">
207 <input type="email" value="" placeholder="Email" name="email" required>
208 </div>
209 <div class="form-group">
210 <input type="text" value="" placeholder="Phone" name="phone" required>
211 </div>
212 <div class="form-group">
213 <input type="text" value="" placeholder="Address" name="address" required>
214 </div>
215 <div class="form-group">
216 <input type="text" value="" placeholder="City" name="city" required>
217 </div>
218 <div class="form-group">
219 <input type="text" value="" placeholder="State" name="state" required>
220 </div>
221 <div class="form-group">
222 <input type="text" value="" placeholder="Zip" name="zip" required>
223 </div>
224 <div class="form-group">
225 <input type="text" value="" placeholder="Subject" name="subject" required>
226 </div>
227 <div class="form-group">
228 <input type="text" value="" placeholder="Message" name="message" required>
229 </div>
230 <input type="button" value="Send" />
231 </div>
```

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <meta charset="UTF-8">
5 <title>Web Phishing Detection site</title>
6 </head>
7 <body>
8 <div class="nav">
9 <a href="#">Home</a>
10 <a href="#">About</a>
11 <a href="#">Contact</a>
12 </div>
13 <div class="about">
14 <h1>Web Phishing Detection site</h1>
15 <p>Web Phishing is one of many security threats to web services on the Internet.
16 Web phishing aims to steal private information, such as usernames, passwords, and credit card
17 details, by way of impersonating a legitimate entity.
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19 <p>Web Phishing is one of many security threats to web services on the Internet.
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30 <input type="text" value="" placeholder="Address" name="address" required>
31 <input type="text" value="" placeholder="City" name="city" required>
32 <input type="text" value="" placeholder="State" name="state" required>
33 <input type="text" value="" placeholder="Zip" name="zip" required>
34 <input type="text" value="" placeholder="Subject" name="subject" required>
35 <input type="text" value="" placeholder="Message" name="message" required>
36 <input type="button" value="Send" />
37 </form>
38 </div>
39 </body>
40 </html>
```

```
304 <div class="contact-list">
305 <div class="list-item">
306 <div class="fa fa-map-marker fa-2x"></div>
307 <div class="contact-text">
308 <div class="text">TAMILNADU, INDIA</div>
309 </div>
310 <div class="list-item">
311 <div class="fa fa-phone fa-2x"></div>
312 <div class="contact-text">
313 <div class="text">222-555-5555</div>
314 </div>
315 <div class="list-item">
316 <div class="fa fa-envelope fa-2x"></div>
317 <div class="contact-text">
318 <div class="text">mailto:info@tamildetection.com</div>
319 </div>
320 </div>
321 </div>
```

4.Contact Page.html

CHAPTER 8

TESTING

8.1 TEST CASES

For the URL verifier module in the ISOT phishing detection system, phishing detection is done using 16 different heuristic rules. In the system, 11 main classes were defined, and 1 class was defined with 5 sub-classes. This covers all 16 heuristic rules. To test the system, 15 test cases were designed using assertion methods. Ten test cases were designed to test the 10 main classes and 5 test cases were designed to test the class with five sub-classes. The getter-setter method was used to test the class with five sub-classes. The getter method is used to obtain or retrieve a variable value from the class, and the setter method is used to store the variables.

The class with five sub-classes checks the 5 different heuristic rules, length of the URL, number of dots and slashes in the URL, presence of @ symbols in the URL, IP address mentioned in the URL, and the presence of special character such as ',', '_', ';' in the URL. Initially, only a single test case was created for the class with five sub-classes, but it was failing as this class has five methods. After applying the getter setter method, all the test cases passed without any issues. The test results are assert Not Null() is used to check if the input URL is not empty, and assert Array Equals() is used to compare the result from the detection method with the expected result.

8.2 USER ACCEPTANCE TESTING

User Acceptance Testing (UAT) is a type of testing performed by the end user or the client to verify/accept the software system before moving the software application to the production environment. The main Purpose of UAT is to validate end to end business flow. It does not focus on cosmetic errors, spelling mistakes or system testing. User Acceptance Testing is carried out in a separate testing environment with production-like data setup. It is kind of black box testing where two or more end-users will be involved.

UAT is performed by –

- Client
- End users

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	9	2	5	2	18
Duplicate	1	0	3	0	4
External	2	2	0	1	5
Fixed	1	2	4	14	21
Not Reproduced	1	0	1	0	2
Skipped	2	0	1	1	4
Won't Fix	2	2	2	1	7
Totals	18	8	16	19	61

Defect Analysis

Need of User Acceptance Testing arises once software has undergone Unit, Integration and System testing because developers might have built software based on requirements document by their own understanding and further required changes during development may not be effectively communicated to them, so for testing whether the final product is accepted by client/end-user, user acceptance testing is needed.

Section	Total Cases	Not Tested	Fail	Pass
Print Engine	5	0	1	4
Client Application	20	0	5	15
Security	2	0	1	1
Outsource Shipping	3	0	0	3
Exception Reporting	7	0	0	7
Final Report Output	3	0	0	3
Version Control	1	0	0	1

Test Case Analysis

CHAPTER 9

RESULT

9.1 PERFORMANCE METRICES

In a classification problem, the category or classes of data is identified based on training data. The model learns from the given dataset and then classifies the new data into classes or groups based on the training. It predicts class labels as the output, such as Yes or No, 0 or 1, Spam or Not Spam, etc. To evaluate the performance of a classification model, different metrics are used, and some of them are as follows:

- Accuracy
- Confusion Matrix
- Precision

S.No.	Parameter	Values	Screenshot																														
1.	Metrics	Regression Model: K-Nearest Neighbors Classification Model: • Confusion Matrix - $\begin{bmatrix} 928 & 50 \\ 86 & 1147 \end{bmatrix}$ • Accuracy Score - 93.848937 • Recall Score - 93.025142 • ROC Score - 93.956334 • Classification Report: precision - 0.94 recall - 0.94 f1 - score - 0.94 support - 2211	• Model building <pre>In [30]: from sklearn.neighbors import KNeighborsClassifier error = [] for i in range(1, 30): knn = KNeighborsClassifier(n_neighbors=i) knn.fit(train_x, train_y) pred_i = knn.predict(test) error.append(np.mean(pred_i != y_test)) In [31]: knn = KNeighborsClassifier(n_neighbors=5) model_kn = knn.fit(train_x, train_y) In [32]: knn.predict(model_kn.predict(test))</pre> • Check the metrics of the model <pre>In [33]: from sklearn.metrics import accuracy_score, recall_score, f1_score, confusion_matrix print("Accuracy Score: %f" % (accuracy_score(y_pred_kn, test))) print("Recall Score: %f" % (recall_score(y_pred_kn, test))) print("ROC Score: %f" % (roc_auc_score(y_pred_kn, test))) print(confusion_matrix(y_pred_kn, test)) Accuracy Score: 0.93848937 Recall Score: 0.93025142 ROC Score: 0.93956334 [[928 50] [86 1147]]</pre> • Classification Report <pre>In [34]: print(classification_report(y_pred_kn, test))</pre> <table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>-1</td><td>0.93</td><td>0.93</td><td>0.93</td><td>978</td></tr><tr><td>1</td><td>0.95</td><td>0.95</td><td>0.94</td><td>1233</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.94</td><td>2211</td></tr><tr><td>macro avg</td><td>0.94</td><td>0.94</td><td>0.94</td><td>2211</td></tr><tr><td>weighted avg</td><td>0.94</td><td>0.94</td><td>0.94</td><td>2211</td></tr></tbody></table>		precision	recall	f1-score	support	-1	0.93	0.93	0.93	978	1	0.95	0.95	0.94	1233	accuracy			0.94	2211	macro avg	0.94	0.94	0.94	2211	weighted avg	0.94	0.94	0.94	2211
	precision	recall	f1-score	support																													
-1	0.93	0.93	0.93	978																													
1	0.95	0.95	0.94	1233																													
accuracy			0.94	2211																													
macro avg	0.94	0.94	0.94	2211																													
weighted avg	0.94	0.94	0.94	2211																													

Metrices [Both Regression Model & Classification Model]

2.	Tune the Model	Hyperparameter Tuning - (GridSearchCV) clf.best_score_ - 0.9315919723544503 Validation Method - GridSearchCV (estimator=SVC())	<pre> In [37]: from sklearn.svm import SVC model = SVC() In [38]: model.fit(x_train,y_train) Out[38]: SVC() In [39]: y_pred = model.predict(x_test) y_pred=model.predict(x_train) • Hyperparameter Tuning GridSearchCV In [40]: from sklearn.model_selection import GridSearchCV In [41]: parameters = {'kernel':('linear','rbf'), 'C':(0.1,1,10), 'gamma':('scale','auto')} In [42]: clf=GridSearchCV(SVC(),param_grid=parameters,cv=5) In [43]: clf.fit(x_train,y_train) Fitting 5 folds for each of 12 candidates, totalling 60 fits [0] BD0.61, gamma:scale, kernel:rbf; total time: 6.3s [1] BD0.61, gamma:scale, kernel:rbf; total time: 6.7s [2] BD0.61, gamma:scale, kernel:rbf; total time: 6.7s </pre>
----	----------------	---	---

Metrics [Tune the Model]

Testcases Report Template - Excel														
chandravarama@gmail.com														
File Home Insert Page Layout Formulas Data Review View Help Tell me what you want to do														
B10														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1					Date	18-Nov-22								
2					Team ID	PNT2022TMD02210								
3					Project Name	Project - Web Phishing Detection								
4					Maximum Marks	4 marks								
5	Test case ID	Feature Type	Component	Test Scenario	Pre-Requisite	Steps To Execute	Test Data	Expected Result	Actual Result	Status	Comments	TC for Automation(Y/N)	BUG ID	Executed By
6	Project_TC_D01	Functional	Home Page	Verify user to enter into the Home page		1.Click the link to open the Application 2. Click on Home page to visit the Home site of the Application	http://127.90.20.5000	Home page should display	Working as expected	Pass	Simple User to open the project	Y		Sharath J R, SathishKumar C
7	Project_TC_D02	UI	Predict Page	Verify the UI element such as predict page		1.Enter URL and click Home 2. Click on Get_started button to predict 3.Verify Predict page with required UI elements	http://127.90.20.5000/home	Application should show below UI elements: a. Get_started button b. watch video c. About Button d. Contact Button	Working as expected	pass	Steps are clear and following elements are verified	Y		Chandru G.Sathish Kumar C
8	Project_TC_D03	Functional	Predict Page	Verify user is able to navigate all sections in the Application		1. Enter URL(https://google.com/) and click predict 2. Click and wait for sometime to predict 3. It shows a Legitimate site and also displays url of the specific site which is searched 4. Click the site which is predicted one. 5. Click on home button to visit home page	https://127.90.20.5000/newdict	User should navigate to homepage and predictpage in every sections	Working as expected	pass	It shows all aspects of the project structure and behaviour	Y		Sharath J R, Ajayeswaran R
9	Project_TC_D04	Functional	Predict Page	Verify user is able to predict the URL is phishing or not		1. Enter URL(http://bet810.com/) and click predict 2. Click and wait for sometime to predict 3. It shows a Phishing site and also displays url of the specific site which is searched 4. Click the site which is predicted one. 5. Click on home button to visit home page	http://127.90.20.5000/newdict	Application should show 'url' and valid message	Working as expected	pass	It shows how it predict and gives accuracy results along with url	Y		Chandru G.Ajeyeswaran R
10														
11														
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13														
14														
15														
16														
17														
18														
19														
20														
Web Phishing Testcases														

Test Cases Report

CHAPTER 10

ADVANTAGES & DISADVANTAGES

ADVANTAGES:

- Using this Detection tool, we easily predict the phishing or legitimate site.
- By using the Machine learning Technique, we can predict and get accuracy-based results.
- In this detection Application, we can predict any URL which makes suspicious.
- It shows URL link below the prediction results, hence it can be redirect easily.
- It shows color indication which clearly defines whether it is phishing or legitimate.
- It is a Machine Learning project; hence it keeps updating on side-by-side.
- By creating this Application using flask, Easy to use and explore everywhere.
- It detects popups, internal links, port, https or http to conclude the Results.
- Now-a-days, Many Applications is developed based on Security based and phishing detection tool, it also plays vital role to detect such phishing URL.

DISADVANTAGES:

- Security becomes doubtful, if such sites can't be predicted by using this site.
- URL format will only be allowed whereas text format is not allowed.
- It is a machine learning Application; dataset sometimes causes wrong results.
- Sometimes, SQL Injection attacks may be occurring on the text area.

CHAPTER 11

CONCLUSION

Overall, phishing is real and dangerous. Everyone needs to watch out for it because it happens to everybody; and getting scammed can be costly. Thus, be careful, follow this advice; and always keep your eyes and ears open to anything that even remotely sounds off because it may very well be scammers looking for their next victim.

Phishing campaigns can be difficult spot. Cyber criminals have become experts at using sophisticated techniques to trick victims into sharing personal or financial information. From eavesdropping to mail tampering, criminals have always sought to steal information as a precursor to launching other exploits.

Phishing websites are being designed to trick people into submitting credentials to access private information and assets by making the sites look like legitimate websites. Phishing attacks through URLs embedded in emails or SMS messages are one of the major issues faced by Internet users because of the huge number of online transactions performed daily. Phishing attacks cause losses to organizations, customers and Internet users.

As it has always been, each individual must shoulder the responsibility to protect themselves from trickery and deception. There are software tools, such as spam filters and antivirus software, that can help, but in the end, we must all be ever-diligent and even a little suspicious of email and SMS communications.

We achieved 93.84% detection accuracy using K-Nearest Neighbours algorithm with lowest false positive rate. Also result shows that classifiers give better performance when we used more data as training data. This paper aims to enhance detection method to detect phishing websites using machine learning technology. We achieved 93.84% detection accuracy using K-Nearest Neighbours algorithm with lowest false positive rate. Also result shows that classifiers give better performance when we used more data as training data. In future hybrid technology will be implemented to detect phishing websites more accurately, for which K-Nearest Neighbours Algorithm of machine learning technology and blacklist method will be used.

CHAPTER 12

FUTURE SCOPE

Phishing is a considerable problem differs from the other security threats such as intrusions and Malware which are based on the technical security holes of the network systems. The weakness point of any network system is its Users. Phishing attacks are targeting these users depending on the trikes of social engineering. Despite there are several ways to carry out these attacks, unfortunately the current phishing detection techniques cover some attack vectors like email and fake websites. This paper develops detection system with a wide protection scope using URL features only which is relying on the fact that users directly deal with URLs to surf the internet and provides a good approach to detect malicious URLs as proved by previous studies. This is important if more heuristics rules are included in the detection system. If the URL length is very long i.e. more than a million characters, then the system may crash. To prevent this situation, a timeout feature can be added when determining the URL length.

In future if we get structured dataset of phishing, we can perform phishing detection much faster than any other technique. In future we can use a combination of any other two or more classifier to get maximum accuracy. We also plan to explore various phishing techniques that uses Lexical features, Network based features, Content based features, Webpage based features and HTML and JavaScript features of web pages which can improve the performance of the system. In particular, we extract features from URLs and pass it through the various classifiers

Additionally, Anti-phishing solutions can be positioned at different levels of attack flow where most researchers are focusing on client-side solutions which turn to add more processing overhead at the client side and lead to losing the trust and satisfaction of the users. Nowadays many organizations make centralized protection of spam filtering. This paper proposes a system which can be integrated into such process in order to increase the detection performance in a real time. The simulation results of the proposed system showed a phishing URLs detection accuracy with 93% and provided online process of a single URL in average time of 0.12 second.

CHAPTER 13

APPENDIX

13.1 SOURCE CODE

```
import pickle
import warnings

import numpy as np

warnings.filterwarnings('ignore')

from flask import Flask, redirect, url_for, render_template, request

from inputScript import FeatureExtraction

app = Flask(__name__)
model = pickle.load(open('Phishing_website.pkl', 'rb'))

#Home page for the project
@app.route('/')
def home():
    return render_template('home.html')

#Redirects to the page to give the user input URL
@app.route('/prediction_page')
def get_started():
    return redirect(url_for('predict'))

#Fetches the URL given by the user and passes to inputScript
@app.route('/predict', methods=['GET', 'POST'])
def predict():
    if request.method == 'POST':
        url = request.form['url']
        ob = FeatureExtraction(url)
        z = np.array(ob.getFeaturesList()).reshape(1,30)
        y_pred=model.predict(z)[0]
        c=model.predict_proba(z)[0,0]
        f=model.predict_proba(z)[0,1]
```

```

        if(y_pred==1):
            return render_template('get_started.html',prediction_text="It is
Legitimate safe website",pred=y_pred,url=url)
        else:
            return render_template('get_started.html',prediction_text="It is a
phishing Website",pred=y_pred,url=url)
        else:
            return render_template('get_started.html')

#contact page for the project
@app.route('/contact')
def contact():
    return render_template('contact.html')

#about page for the project
@app.route('/about')
def about():
    return render_template('about.html')

if __name__=='__main__':
    app.run(debug = True)

```

```

<html>
  <head>
    <title>detecting page</title>
    <meta name="viewport" content="width=device-width, initial-scale=1">

  </head>
  <style>
    *{
      box-sizing: border-box;
    }
    body{
      margin: 0;
    }
    .head{
      height: 60px;
      font-size: 40px;
    }
  </style>

```



```

        color: white;
        overflow: hidden;
        background-color: darkslateblue;
    }

<div class = "main">
    <h1 style="color:midnightblue;font-size: 35px">Web Phishing Detection
using Machine Learning</h1>
    <form action="/predict" method='post'>
        <input style="background-color:ivory;border-radius: 5px;width:
50%;height: 30px; color:black; text-align: left; margin: 30px" type="url"
name="url" placeholder="Enter the URL to be verified" required><br>
        <input id="butto" type="submit" value="Predict" name="Predict">
    <div>
        {% if pred == 1 %}
            <p style="color:green;text-decoration: none;font-
size:30px;margin-top: 30px;">{{ prediction_text }}</p>
            <p style="color:blue;font-size: 20px;"><a href="{{ url }}">{{
url }}</a></p>
        {% else %}
            <p style="color:red;text-decoration: none;font-
size:30px;margin-top: 30px;">{{ prediction_text }}</p>
            <p style="color:blue"><a href="{{ url }}">{{ url }}</a></p>
        {% endif %}
    </div>
</form>
</div>
</body>

```

13.2 GITHUB AND PROJECT DEMO LINK

GitHub Link: <https://github.com/IBM-EPBL/IBM-Project-12374-1659448777>

Demo Link: <https://vimeo.com/772851659>